

Lab1 : Apache Web Server

Install Apache HTTP server

```
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$ sudo systemctl status apache2  
[sudo] password for nagy:  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/lib/systemd/system/apache2.service; enabled; vendor preset: enabled)  
   Active: active (running) since Sun 2026-03-08 13:56:05 EET; 5min ago  
     Docs: https://httpd.apache.org/docs/2.4/  
    Main PID: 760673 (apache2)  
      Tasks: 55 (limit: 18720)  
     Memory: 5.9M  
        CPU: 56ms  
    CGroup: /system.slice/apache2.service  
            └─760673 /usr/sbin/apache2 -k start  
              └─760675 /usr/sbin/apache2 -k start  
                └─760676 /usr/sbin/apache2 -k start  
  
Mar 08 13:56:04 ZenBook-UX425EA systemd[1]: Starting The Apache HTTP Server...  
Mar 08 13:56:05 ZenBook-UX425EA systemd[1]: Started The Apache HTTP Server.  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$
```

Create two simple html pages named “page1.html, page2.html” then use the suitable directive to automatically redirect from localhost/page1.html to localhost/page2.html.

```
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$  
nagy@ZenBook-UX425EA:~/Mine/ITI/017_WebServers/Apache2/Lab$ cd /var/www/html  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/page1.html  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/page2.html  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost/page1.html  
<!DOCTYPE html>  
<html>  
<head>  
  <title>Page 1</title>  
</head>  
<body>  
  <h1>Welcome to Page 1</h1>  
</body>  
</html>  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost/page2.html  
<!DOCTYPE html>  
<html>  
<head>  
  <title>Page 2</title>  
</head>  
<body>  
  <h1>Welcome to Page 2</h1>  
</body>  
</html>  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$  
nagy@ZenBook-UX425EA:/var/www/html$
```

Ask for username and password when accessing a directory.

```
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /etc/apache2/sites-available/000-default.conf
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/.htaccess
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo htpasswd -c /etc/apache2/.htpasswd test
New password:
Re-type new password:
Adding password for user test
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo systemctl restart apache2.service
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost/page1.html
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>401 Unauthorized</title>
</head><body>
<h1>Unauthorized</h1>
<p>This server could not verify that you
are authorized to access the document
requested. Either you supplied the wrong
credentials (e.g., bad password), or your
browser doesn't understand how to supply
the credentials required.</p>
<hr>
<address>Apache/2.4.52 (Ubuntu) Server at localhost Port 80</address>
</body></html>
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
```

Create a directory then allow access to one of your classmates only.

```
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /etc/apache2/apache2.conf
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/.htaccess
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo systemctl restart apache2.service
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl -u test:19919690 http://localhost/page1.html
<!DOCTYPE html>
<html>
<head>
<title>Page 1</title>
</head>
<body>
<h1>Welcome to Page 1</h1>
</body>
</html>
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl -u test:19919690 http://localhost/page2.html
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>302 Found</title>
</head><body>
<h1>Found</h1>
<p>The document has moved <a href="http://localhost/page1.html">here</a>.</p>
<hr>
<address>Apache/2.4.52 (Ubuntu) Server at localhost Port 80</address>
</body></html>
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
```

Disable listing the directory content (hint use indexes)

```
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/.htaccess
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /etc/apache2/sites-available/000-default.conf
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo systemctl restart apache2.service
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
  <!--
    Modified from the Debian original for Ubuntu
    Last updated: 2022-03-22
    See: https://launchpad.net/bugs/1966004
  -->
  <head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
    <title>Apache2 Ubuntu Default Page: It works</title>
    <style type="text/css" media="screen">
      * {
        margin: 0px 0px 0px 0px;
        padding: 0px 0px 0px 0px;
      }

      body, html {
        padding: 3px 3px 3px 3px;

        background-color: #D8DBE2;

        font-family: Ubuntu, Verdana, sans-serif;
        font-size: 11pt;
        text-align: center;
      }

      div.main_page {
        position: relative;
        display: table;

        width: 800px;

        margin-bottom: 3px;
        margin-left: auto;
        margin-right: auto;
        padding: 0px 0px 0px 0px;
```

Change the default index page to be default.html instead of index.html

```
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo nano /var/www/html/.htaccess
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ vim /var/www/html/default.html
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ sudo vim /var/www/html/default.html
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$ curl localhost
<!DOCTYPE html>
<html>
<head>
  <title>Default Page</title>
</head>
<body>
  <h1>Welcome to Default Page</h1>
</body>
</html>
nagy@ZenBook-UX425EA:/var/www/html$
nagy@ZenBook-UX425EA:/var/www/html$
```